1.2 Simulation of fracture energy measurement (GFM)

Keywords: plain concrete, softening, prescribed-displacement boundary condition, Newton-Raphson method, mesh refinement

Input files: GFM\GFM1.cc2 (shorter notch)
GFM\GFM2.cc2 (longer notch)

1.2.1 Introduction

Fracture energy of a quasi-brittle material can be measured by conducting two fracture tests on double cantilever beam (DCB) specimens, with identical dimensions but slightly different initial notch lengths a_1 and a_2. Fracture energy of the tested material is then obtained as the area enclosed by the two specimens' load-displacement curves, divided by the initial notch area difference $\Delta a.b$.

$$G_f = \frac{W_1 - W_2}{\Delta a.b} \tag{1.2}$$

In this example, we reproduce this measurement method by FE analyses. The measured fracture energy is compared with the fracture energy, which is input as a material parameter. The tested specimen dimensions and material properties are taken from Okada and Horii [1].

1.2.2 Comments on FE model preparation

1.2.2.1 Topology

The analyzed beam geometry and support and loading conditions are shown in Figure 1-3. The specimen thickness b_0 is 120 mm; however, to ensure a straight crack propagation, the specimen contains on both sides ten-millimeter-deep grooves along the symmetry plane. Consequently, the area weakened by the guides is represented by a separate macroelement with thickness b=100 mm. In order to achieve sufficient accuracy of the analysis, finite element mesh is refined around the expected crack path, so as to discretize the ligament by at least 20 elements.

Two specimens with different initial notch lengths a_1=100 mm and a_2=105 mm are analyzed. Thus the difference, Δa, is equal to 5 mm.

1.2.2.2 Materials

Concrete is modeled by the ‘Sbeta Material’ model. Relevant material properties used in the analysis are taken from Okada and Horii [1] and are listed in Table 1.2-1. The rotating crack model is used.

1.2.2.3 Loads and supports and Solution parameters

Since we need to trace the load-displacement curve up to the post-failure softening regime, we prescribe loading by controlling the displacements at the loading points and employ the Newton-Raphson solution method. The full Newton-Raphson method, with the tangent stiffness updated in each iteration, is used together with the line search technique. Relevant options and parameters as summarized in Table 1.2-2.

1.2.2.4 Analysis steps

Due to the presence of a notch, cracking starts at a very low load. Thus, a fine stepping has to be used from the very beginning of the analysis. The load-point displacement increment is kept constant, 0.003 mm.

1.2.3 Results

Figure 1-4 shows the load-displacement curves for the specimens with different initial notch lengths. The curves are numerically integrated to obtain the energies:

W_1 = 0.935 Nm (shorter notch a_1)

W_2 = 0.873 Nm (longer notch a_2)

With Δa = 5 mm and b = 100 m, Eq. (1.2) yields the value of G_f = 124.591 N/m. This value is only slightly higher than the inputted fracture energy (122 N/m), which proves validity of the measurement method. The slight overestimation results from the fact that cracking takes place not only on the main crack plane (as it is assumed in Eq. (1.2)), but some fine cracks also occur in directions almost perpendicular to the main crack.

G_f computed (N/m)	G_f input (N/m)
124.591	122

1.2.4 References

[1] Okada T. and Horii, H., Effect of Specimen Size and Loading Rate on the Tension-Softening Curve Obtained by Back Analysis Method, in Fracture Mechanics of Concrete Structures – Proc. of FRAMCOS-3 (Reinhardt and Naaman eds.), AEDIFICATIO Publishers, 1998, pp. 89-100

Table 1.2-1 Material properties

Material type		SBETA material	
Elastic modulus	E_c	27.6	GPa
Poisson's ratio	ν	0.21	-
Compressive strength	f_c	26.4	MPa
Tensile strength	f_t	2.06	MPa
Type of tension softening		Exponential	
Fracture energy	G_f	122.0	N/m
Crack model		Rotating	

Table 1.2-2 Solution parameters

Solution method	Newton-Raphson	
Stiffness/update	Tangent/each iteration	
Number of iterations	50	
Error tolerance	0.010	
Line search	on, with iterations	

Table 1.2-3 Finite element mesh

Finite element type	Quadrilateral, CCIsoQuad	
Element shape smoothing	on	
Optimization	Sloan	

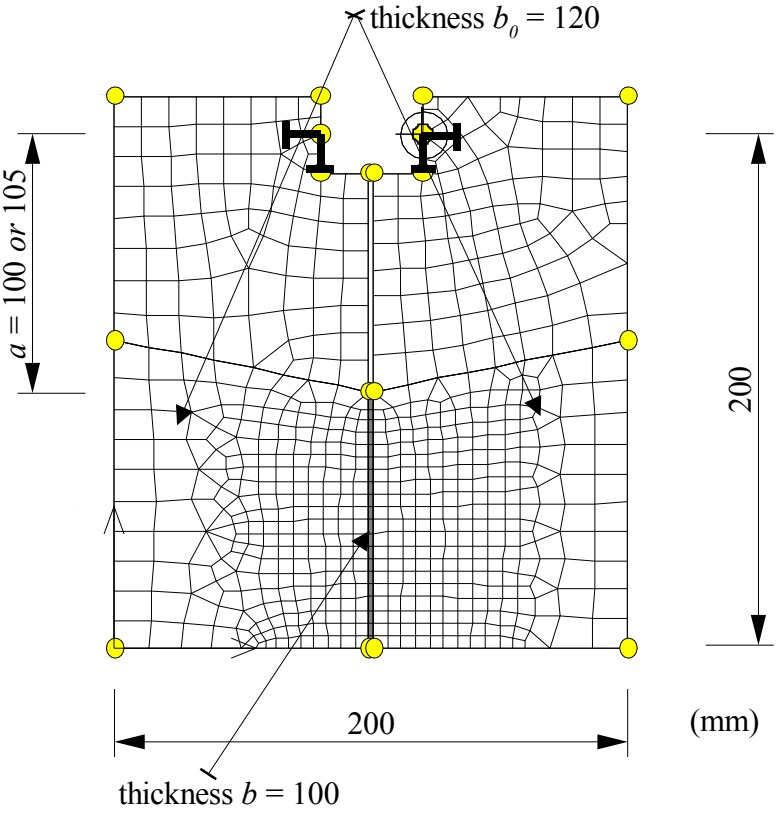


Figure 1-3: Configuration and FE mesh of the double cantilever beam specimen

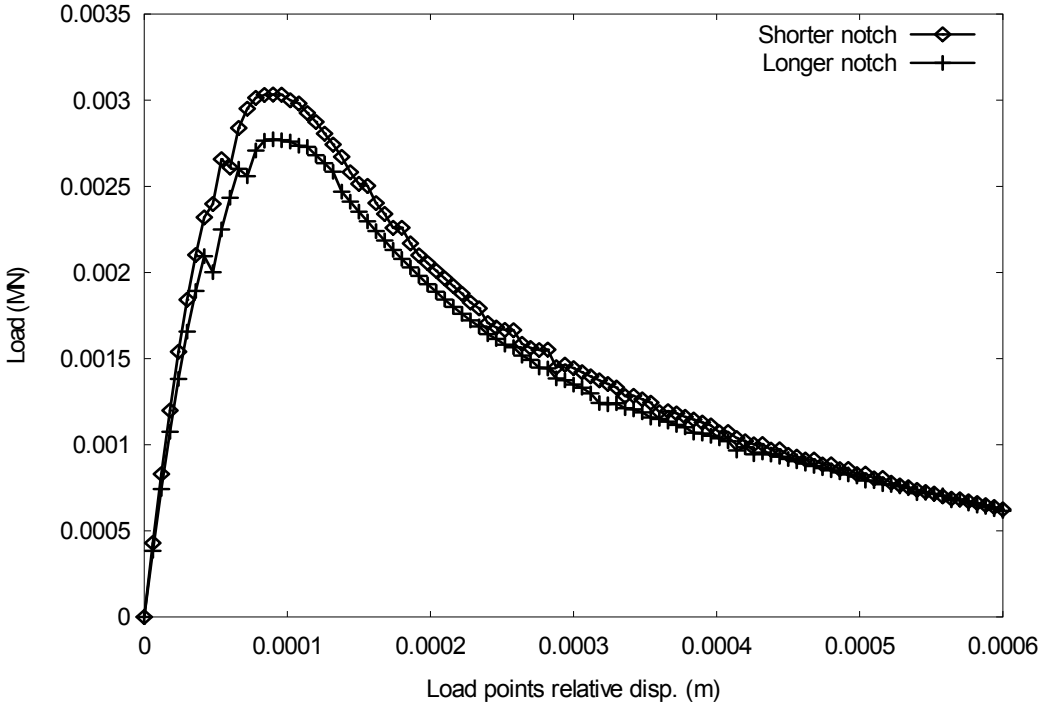


Figure 1-4: Load-displacement curves of DCB specimens